TORTO

Intermediate presentation

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Project goal

A training assistant that observes exercise
without wearables, but with critical condition spotting ability.

Arm-lift exercise

Count repetitions
Estimate arm angles

Measurements

CO_2 and temperature
Humidity and brightness
Body temperature

The current prototype

TurtleBot3 Burger + Pi 4

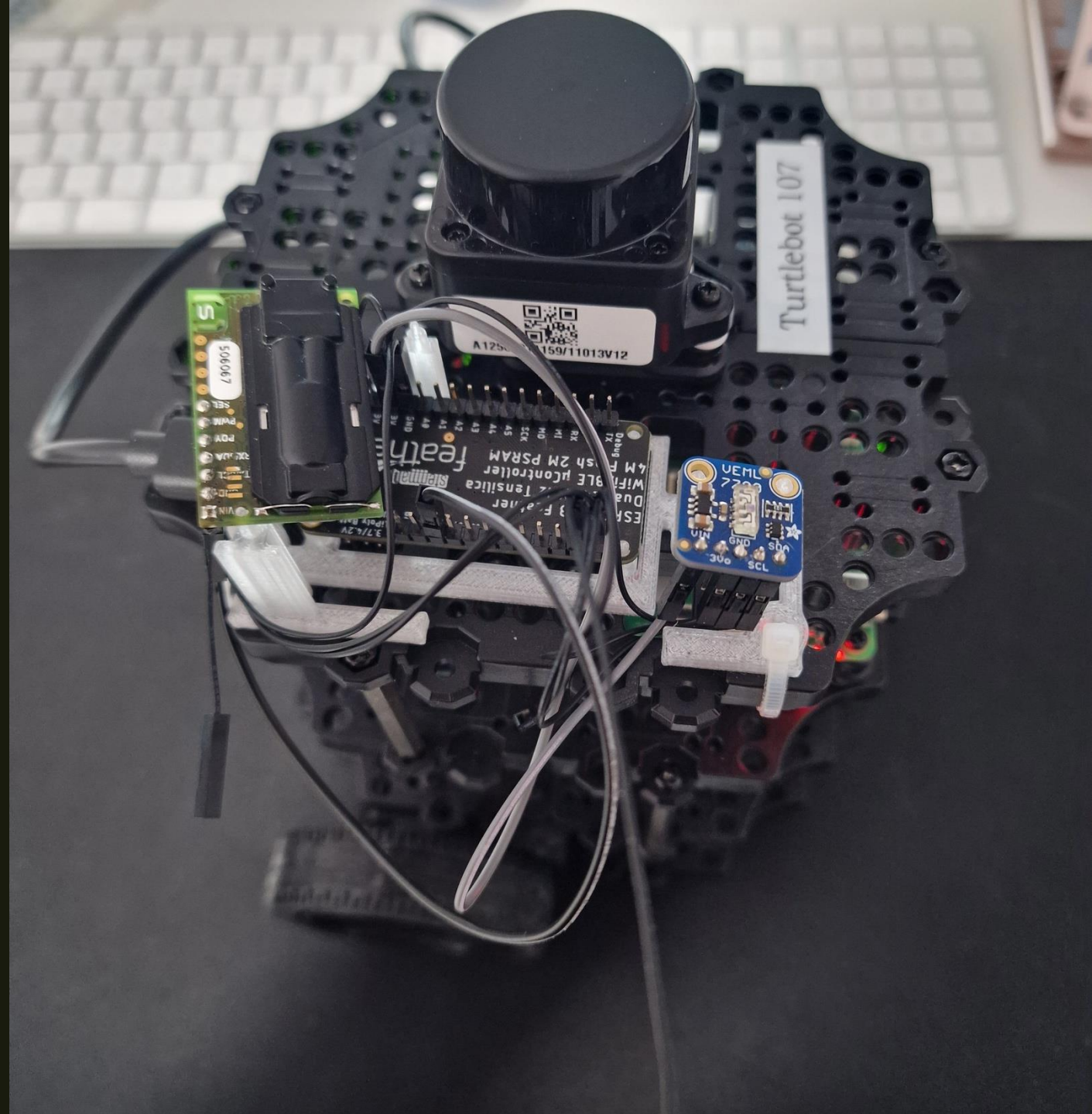
Feather ESP32-S3

SCD30

CO_2, temperature and humidity

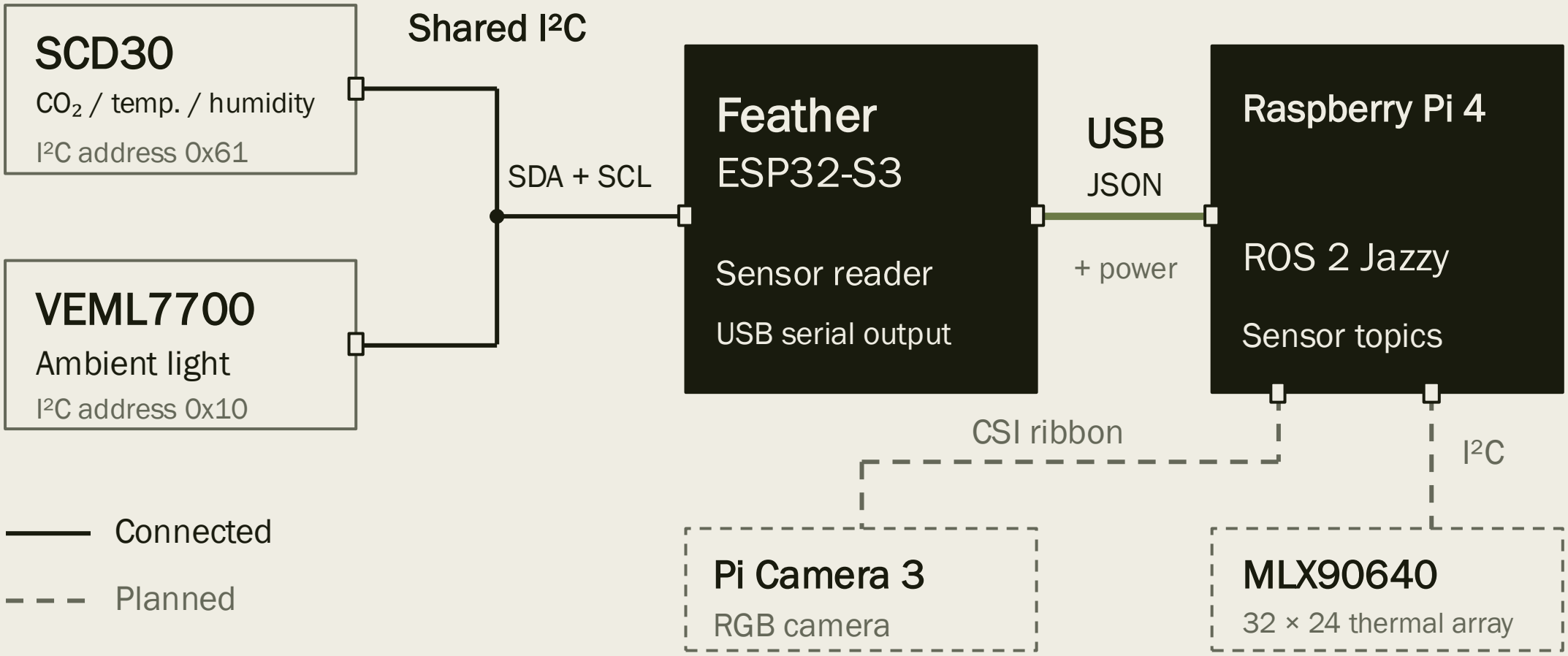
VEML7700

Illuminance (lux)



Sensor and camera connections

Two sensors connected. Camera integration is next.



Layout and camera mounts

Top level reserved for cameras and sensors.

15° upward tilt for cameras, or maybe more or less we will see ;)



Live sensor test

```
Wed Nov  4 09:21:26 PM CET 2026
{
  "passed": true,
  "counts": {
    "co2_ppm": 3,
    "temperature": 3,
    "relative_humidity": 3,
    "illuminance": 3
  },
  "latest": {
    "co2_ppm": 639.47,
    "temperature": 23.697,
    "relative_humidity": 0.38341000000000003,
    "illuminance": 81.818
  },
  "diagnostics": {
    "feather/scd30": {
      "level": 0,
      "message": "Receiving measurements"
    },
    "feather/veml7700": {
      "level": 0,
      "message": "Receiving measurements"
    }
  }
}
ubuntu@tbot:~$
```

Live SSH session, 4 November 2026

TurtleBot 107

CO_2: 639 ppm
Temperature: 23.7 °C
Relative humidity: 38.3%
Illuminance: 81.8 lux

Test passed

3 messages per topic
Both sensors healthy

Software

Implemented

Feather reads both sensors.
USB sends JSON samples.
Pi publishes 4 ROS 2 topics.
Invalid data is somewhat filtered.

Next

Read camera images.
Estimate arm angles.
Count lift-and-return cycles.
Show feedback and thermal view.
Save session results.
Choose correct filming position.
Make some nice endpoint, to see all data from robot easily.

THANK YOU

Questions?

One more thing...

We promised a shell for Torto and he is going to get one.

Stay tuned.

